

DBMS – SQL QUERY IMPLEMENTATION FOR E-COMMERCE DATABASE

Project Title: EcoCart – Sustainable Products E-Commerce Order Management System

Skill Task: DBMS

Week: 7

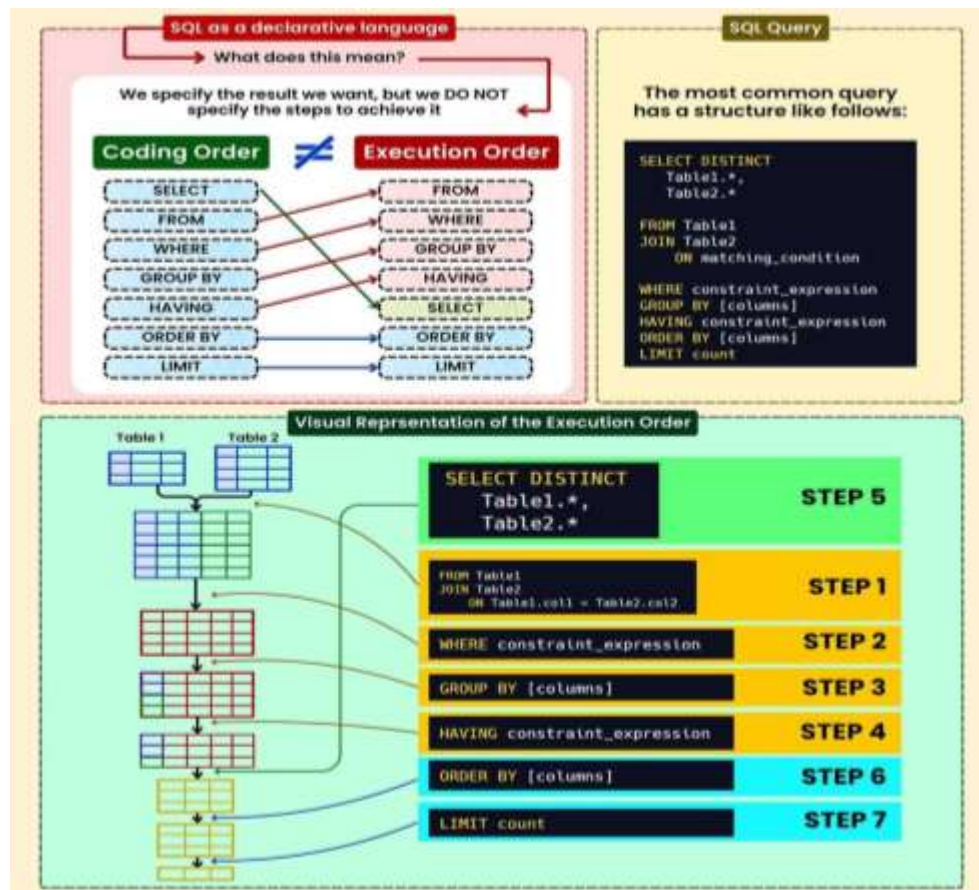
Objectives

1. Perform **SELECT**, **WHERE**, **ORDER BY**, and **DISTINCT** queries.
2. Search products based on price, category, and availability.
3. Retrieve customer and product information.
4. Apply filtering conditions.
5. Generate basic business reports.

1. Basic SQL Queries

1.1 SELECT Query

The SELECT statement is used to retrieve data from a table.



Display all products

```
SQL> SELECT * FROM Product;
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
CATEGORY_ID PRICE STOCK_QUANTITY
```

```
101
Organic Turmeric Powder
1 130 45
```

```
102
Black Pepper
1 180 40
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
CATEGORY_ID PRICE STOCK_QUANTITY
```

```
103
Organic Honey
2 350 25
```

```
104
Bamboo Storage Box
```

Display selected product details

```
SQL> SELECT Product_ID, Product_Name, Price FROM Product;
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
PRICE
```

```
101
Organic Turmeric Powder
130
```

```
102
Black Pepper
180
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
PRICE
```

```
103
Organic Honey
350
```

```
104
```

1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition.

Find products with price less than 1000

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price < 1000;
```

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID	PRICE	STOCK_QUANTITY
101	Organic Turmeric Powder	1	130	45
102	Black Pepper	1	180	40
103	Organic Honey			

Find products with price greater than 100

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price > 100;
```

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID	PRICE	STOCK_QUANTITY
101	Organic Turmeric Powder	1	130	45
102	Black Pepper	1	180	40
103	Organic Honey			

1.3 ORDER BY

The ORDER BY clause is used to sort the query results.

Products from lowest to highest price

```
SQL> SELECT Product_ID, Product_Name, Price
2 FROM Product
3 ORDER BY Price ASC;
```

PRODUCT_ID

PRODUCT_NAME

PRICE

101
Organic Turmeric Powder
130

102
Black Pepper
180

PRODUCT_ID

PRODUCT_NAME

PRICE

103
Organic Honey

Products from highest to lowest price

```
SQL> SELECT Product_ID, Product_Name, Price
2 FROM Product
3 ORDER BY Price DESC;
```

PRODUCT_ID

PRODUCT_NAME

PRICE

104
Bamboo Storage Box
450

103
Organic Honey
350

PRODUCT_ID

PRODUCT_NAME

PRICE

102
Black Pepper

1.4 DISTINCT

The DISTINCT keyword is used to display unique values.

Display unique categories

```
SQL> SELECT Product_ID, Product_Name, Price
2 FROM Product
3 ORDER BY Price DESC;
```

PRODUCT_ID

PRODUCT_NAME

PRICE

104
Bamboo Storage Box
450

103
Organic Honey
350

PRODUCT_ID

PRODUCT_NAME

PRICE

2. Search Products

2.1 Search Products Based on Price

```
SQL> SELECT Product_ID, Product_Name, Price
2 FROM Product
3 WHERE Price BETWEEN 100 AND 400;
```

PRODUCT_ID

PRODUCT_NAME

PRICE

101
Organic Turmeric Powder
130

102
Black Pepper
180

PRODUCT_ID

PRODUCT_NAME

PRICE

103
Organic Honey
350

This displays products whose price is between ₹100 and ₹400.

2.2 Search Products Based on Category

```
SQL> SELECT Product_ID, Product_Name, Category_ID
  2  FROM Product
  3  WHERE Category_ID = 1;
```

PRODUCT_ID

PRODUCT_NAME

CATEGORY_ID

101
Organic Turmeric Powder
1

102
Black Pepper
1

PRODUCT_ID

PRODUCT_NAME

CATEGORY_ID

This displays products belonging to category 101.

3. Retrieve Customer Information

Display all customers

```
SQL> SELECT Customer_ID, Customer_Name, Email, Contact_No
  2  FROM Customer;
```

CUSTOMER_ID

CUSTOMER_NAME

EMAIL

CONTACT_NO

101
Anitha Kumar
anitha@gmail.com
9876501234

CUSTOMER_ID

CUSTOMER_NAME

EMAIL

CONTACT_NO

102
Rahul

Search for a particular customer

```
SQL> SELECT *  
2 FROM Customer  
3 WHERE Customer_ID = 101;
```

CUSTOMER_ID

CUSTOMER_NAME

EMAIL

CONTACT_NO

ADDRESS

REGISTRAT

101

CUSTOMER_ID

CUSTOMER_NAME

EMAIL

CONTACT_NO

ADDRESS

4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT.

4.1 AND Condition

```
SQL> SELECT Product_ID, Product_Name, Price  
2 FROM Product  
3 WHERE Price < 2000  
4 AND Category_ID = 2;
```

PRODUCT_ID

PRODUCT_NAME

PRICE

103

Organic Honey

350

This displays products that satisfy **both conditions**.

4.2 OR Condition

```
SQL> SELECT Product_ID, Product_Name, Price
2 FROM Product
3 WHERE Category_ID = 1
4 OR Category_ID = 3;
```

PRODUCT_ID

PRODUCT_NAME

PRICE

101
Organic Turmeric Powder
130

102
Black Pepper
180

PRODUCT_ID

PRODUCT_NAME

PRICE

This displays products belonging to either category 101 or 102.

4.3 IN Operator

```
SQL> SELECT Product_ID, Product_Name, Price
2 FROM Product
3 WHERE Category_ID IN (1, 2, 3);
```

PRODUCT_ID

PRODUCT_NAME

PRICE

101
Organic Turmeric Powder
130

102
Black Pepper
180

PRODUCT_ID

PRODUCT_NAME

4.4 BETWEEN Operator

```
SQL> SELECT Product_ID, Product_Name, Price
2 FROM Product
3 WHERE Price BETWEEN 100 AND 300;
```

PRODUCT_ID

PRODUCT_NAME

PRICE

101
Organic Turmeric Powder
130

102
Black Pepper
180

PRODUCT_ID

PRODUCT_NAME

PRICE

5. Basic Business Reports

5.1 Product Price Report

```
SQL> SELECT
2 Product_ID,
3 Product_Name,
4 Price
5 FROM Product
6 ORDER BY Price DESC;
```

PRODUCT_ID

PRODUCT_NAME

PRICE

104
Bamboo Storage Box
450

103
Organic Honey
350

PRODUCT_ID

PRODUCT_NAME

PRICE

102
Black Pepper

5.2 Customer Order Report

```
SQL> SELECT
2     c.Customer_ID,
3     c.Customer_Name,
4     o.Order_ID,
5     o.Order_Date,
6     o.Total_Amount
7 FROM Customer c
8 JOIN Orders o
9     ON c.Customer_ID = o.Customer_ID
10 ORDER BY o.Order_Date DESC;
```

CUSTOMER_ID

CUSTOMER_NAME

ORDER_ID	ORDER_DAT	TOTAL_AMOUNT
101		
Anitha Kumar	201 16-SEP-26	2700
102		
Rahul	202 15-SEP-26	1800

CUSTOMER_ID

CUSTOMER_NAME

5.3 Product Order Report

```
SQL> SELECT
2     od.Order_ID,
3     od.Product_ID,
4     od.Quantity,
5     od.Unit_Price,
6     od.Subtotal
7 FROM Order_Details od
8 ORDER BY od.Subtotal DESC;
```

ORDER_ID	PRODUCT_ID	QUANTITY	UNIT_PRICE	SUBTOTAL
203	104	2	1600	3200
201	101	3	1000	3000
202	103	2	900	1800
204	101	1	1500	1500
201	102	1	500	500

5.4 Customer-wise Total Orders

```
SQL> SELECT
  2     c.Customer_ID,
  3     c.Customer_Name,
  4     COUNT(o.Order_ID) AS Total_Orders
  5 FROM Customer c
  6 JOIN Orders o
  7     ON c.Customer_ID = o.Customer_ID
  8 GROUP BY c.Customer_ID, c.Customer_Name
  9 ORDER BY Total_Orders DESC;
```

CUSTOMER_ID	CUSTOMER_NAME	TOTAL_ORDERS
103	Priya	1
102	Rahul	1

5.5 Customer-wise Total Purchase Amount

```
SQL> SELECT
  2     c.Customer_ID,
  3     c.Customer_Name,
  4     SUM(o.Total_Amount) AS Total_Purchase
  5 FROM Customer c
  6 JOIN Orders o
  7     ON c.Customer_ID = o.Customer_ID
  8 GROUP BY c.Customer_ID, c.Customer_Name
  9 ORDER BY Total_Purchase DESC;
```

CUSTOMER_ID	CUSTOMER_NAME	TOTAL_PURCHASE
103	Priya	3200
101	Anitha Kumar	2700